

Chapter 7 — Genitourinary Tract Imaging: Basic Principles of Nuclear Medicine

INTRODUCTION

- **Nuclear medicine** uses radionuclide-labelled pharmaceuticals to **visualise organ function, treat disease and characterise molecular processes within cells**; the last is termed **molecular imaging**.
- **Radiotracers are given in subpharmacological quantities**, so they do not perturb the process being measured. Their emissions are detected by an external sensor and formatted as an image.
- **Contrast with anatomic imaging**: plain radiography, computed tomography (CT), magnetic resonance imaging (MRI) and ultrasonography all need an **external energy source**; in nuclear medicine the patient is the source.

Concept — Nuclear imaging shows what tissue does rather than what it looks like, because the signal comes from a tracer taken up by a physiological or molecular process.

PRINCIPLES OF SINGLE-PHOTON AND PET IMAGING

Single-photon imaging

- **Gamma photons of 100–300 keV** are detected by a **gamma camera**, in planar or tomographic mode.
- **Sequence**: gamma photon → **scintillation crystal** (usually sodium iodide) absorbs it and emits light → **photomultiplier tubes** collect and amplify the light → **circuitry** integrates the output → image.
- **Collimator** (between patient and crystal): its geometry admits photons travelling only in a certain direction, **encoding spatial information**.
- **Planar mode**: detection at a single angle → **two-dimensional** image.
- **Single-photon emission computed tomography (SPECT)**: the camera rotates slowly around the patient → multiple angles → **three-dimensional** image. A **non-contrast CT** is acquired at the same time for **anatomic localisation and attenuation correction** (signal from deeper tissue is attenuated on its way out). SPECT is either routine or a **problem-solving** adjunct when planar images are inadequate.
- **Radionuclides**: technetium-99m (^{99m}Tc) and indium-111 (^{111}In) (Table 7.1).

Positron emission tomography (PET)

- **Sequence**: positron emitted → travels a short distance → collides with an electron → **annihilation** → **two 511-keV photons 180° apart** → **coincidence detection** → image.
- Acquired tomographically, usually with **non-contrast CT** (localisation and attenuation correction), or with MRI in **PET/MRI** scanners at large centres.
- **Radionuclides**: carbon-11 (^{11}C), iodine-124 (^{124}I), fluorine-18 (^{18}F), gallium-68 (^{68}Ga) (Table 7.2).
- **Conspicuity**: PET rejects photons without a coincident partner, minimising noise; a gamma camera treats **any photon through the collimator as true signal**.
- **Quantification**: PET measures **standardised uptake values (SUVs)**, corrected for injected dose and body mass; clinical single-photon imaging is only **semiquantitative**.
- **Limitation of PET**: high cost, limiting access in much of the world.
- **Current practice**: most urological **functional** imaging uses scintigraphy; **cancer** imaging mostly uses PET with CT or MRI.

SCINTIGRAPHY, SPECT AND PET COMPARED

	Planar scintigraphy	SPECT	PET
Detected emission	Single gamma photon	Single gamma photon	Two 511-keV photons in coincidence
Image	2D	3D (rotating camera)	3D
Spatial resolution	~20 mm	~10 mm	~5 mm (best)
Noise handling	Collimator only	Collimator only	Coincidence filtering
Quantitation	Semiquantitative	Semiquantitative	SUV
Cost	Low	Moderate	High
Urological role	Most functional imaging	Problem-solving, cortical imaging	Cancer imaging

TABLE 7.1 PHYSICAL CHARACTERISTICS OF SINGLE-PHOTON EMITTING RADIONUCLIDES MENTIONED IN THE CHAPTER

Radionuclide	Half-life	Principal photon energies, keV (abundance)
^{99m} Tc	6.0 h	140 (89%)
¹¹¹ In	67.3 h	171 (90%), 245 (94%)
Gallium-67 (⁶⁷ Ga)	78.3 h	93 (37%), 185 (20%), 300 (17%), 395 (5%)

Data from Ziessman HA, O'Malley JP, Thrall JH: Nuclear medicine: the requisites, 2013.

- **What the table is for:** half-life sets the imaging window. ^{99m}Tc's 6-hour half-life and single clean 140-keV photon make it the workhorse; ¹¹¹In and ⁶⁷Ga last days, which suits delayed infection imaging.

TABLE 7.2 PHYSICAL CHARACTERISTICS OF POSITRON-EMITTING RADIONUCLIDES MENTIONED IN THE CHAPTER

Radionuclide	Half-life	Positron energy (keV)	Range in soft tissue (mm)	Production
¹¹ C	20.3 min	960	3.9	Cyclotron
¹⁸ F	109.8 min	634 (lowest)	2.3 (shortest)	Cyclotron
⁶⁸ Ga	68.0 min	1,899	8.9	Generator
¹²⁴ I	4.17 days	1,525 and 2,138	6.9 and 10.2	Cyclotron
Zirconium-89 (⁸⁹ Zr)	78.4 h	896	3.6	Cyclotron

Data from Serdons K, Verbruggen A, Bormans GM: Developing new molecular imaging probes for PET. Methods 2009.

- **Pattern:** higher positron energy → longer range before annihilation → blurrier image. ¹⁸F has the **lowest energy and shortest range**, hence superior image quality, plus a half-life long enough for distribution.
- ¹¹C's 20-minute half-life demands an on-site cyclotron; ⁶⁸Ga comes from a generator; ¹²⁴I and ⁸⁹Zr last days, matching the slow kinetics of antibodies.

Concept — PET outperforms single-photon imaging because coincidence detection rejects noise and permits true quantitation, while the choice of radionuclide is a trade between half-life, positron range and logistics.

FUNCTIONAL IMAGING OF THE KIDNEYS

- **Applications:** renal blood flow, **differential renal function**, presence and degree of **obstruction**, and **renal scarring**. Interpretation depends on renal physiology and the properties of each agent.

Relevant Renal Physiology

- **Kidneys receive 20% of cardiac output:** 5 L/min → about **1 L/min** renal blood flow. Plasma is 60% of blood, so **renal plasma flow ≈ 600 mL/min**.
- **Plasma clearance:** about **20% by passive glomerular filtration**, **80% by active tubular secretion**. The kidneys reclaim free water and electrolytes according to need.
- **Principal tracers:** technetium-99m diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA), technetium-99m mercaptoacetyltriglycine (^{99m}Tc-MAG3) and technetium-99m dimercaptosuccinic acid (^{99m}Tc-DMSA).

^{99m}Tc-DTPA

- DTPA (pentetic acid) is a heavy-metal chelator. **Extracted entirely by glomerular filtration**, not reabsorbed → **measures glomerular filtration rate (GFR) directly**.
- Used for differential function and obstruction, but **impaired renal function gives high background and poor images**.

^{99m}Tc-MAG3

- **Protein-bound; cleared by tubular secretion** → **unaffected by impaired GFR**, higher extraction efficiency, better images, **lower non-target radiation**, and **cheaper** than DTPA.
- **Agent of choice for split function and functional obstruction**.

^{99m}Tc-DMSA

- Cleared mainly by **glomerular filtration**, then taken up by **proximal tubular cells by receptor-mediated endocytosis via megalin and cubilin** → **retained in the cortex**.
- **Ideal for cortical processes:** acute pyelonephritis and renal scarring.

RENAL RADIOTRACERS COMPARED

	^{99m} Tc-DTPA	^{99m} Tc-MAG3	^{99m} Tc-DMSA
Handling	Glomerular filtration only	Tubular secretion (protein-bound)	Filtration → proximal tubular retention (megalin, cubilin)
Study type	Dynamic	Dynamic	Static cortical (2 h)
Key use	GFR; renovascular hypertension	Split function, obstruction (first choice)	Pyelonephritis, scarring
In renal impairment	Poor images, high background	Good images	Imaging time may need extending
Cost	Higher	Lower	—

Dynamic Renal Imaging With ^{99m}Tc-MAG3 and ^{99m}Tc-DTPA

- Split function and obstruction require **dynamic imaging over time**, unlike the single static image of most anatomic modalities.

Patient Preparation

- **Good hydration** ensures optimal tracer delivery.
- **Record medications**, particularly diuretics, and any **anatomic abnormalities** affecting positioning, camera set-up or interpretation.
- **Bladder outlet obstruction or neurogenic bladder** → **Foley catheter** to reduce retrograde pressure on the kidneys.
- **Nephrostomy tubes are usually capped** so that native ureteric patency can be tested.

Dosing and Pharmacokinetics

- **5–10 mCi (185–370 MBq) intravenous push** for both MAG3 and DTPA.
- **MAG3 sequence:** extraction nearly 60% → **peak cortical uptake 3–5 min** → collecting system shortly after → **bladder by 10–15 min**. The half-clearance time (peak to 50% clearance) for a non-obstructed unit is given as 15–20 min.

Image Acquisition and Interpretation

- **Perfusion phase:** frames every **1–2 seconds** for about **1 minute**; regions of interest (ROIs) over the aorta, each kidney, and just outside each kidney for **background subtraction**; counts per unit time plotted as a **time–activity curve (TAC)**.
 - Renal activity should appear **within seconds of the aorta**, with a **brisk upstroke matching the aortic curve**.
 - **A slow rise to peak suggests poor flow and likely poor function.**
- **Functional phase:** about **1 frame per minute**. **Relative function = area under each TAC between 1 and 3 minutes**. A **split difference up to 10% is normal** (i.e. up to 45:55).
- **Obstruction:** a healthy kidney clears tracer **within 15 minutes**; an obstructed unit retains it, giving a **progressively rising TAC with no peak**.
- **Delayed pelvic clearance without true obstruction** (a patulous system) is separated by **furosemide: 0.5 mg/kg intravenously, maximum 40 mg in adults**; higher doses in renal impairment.
- **Timing is not agreed:** the authors give furosemide **15 minutes after the tracer (F+15)**; others at the start (F0) or 15 minutes before (F–15). **Always read the report to learn whether and when a diuretic was given.**

DIURETIC RENOGRAPHY — POST-FUROSEMIDE HALF-CLEARANCE TIME

Half-clearance ($T_{1/2}$)	Interpretation
<10 min	Non-obstructed (patulous system)
10–20 min	Indeterminate — further evaluation
>20 min	Obstruction

Figure legends

- **Figure 7.1 — normal MAG3 renogram:** sharp upstroke of aortic and renal curves, aorta preceding kidneys by seconds; renal peak at 3–4 min, half-peak 6–9 min later; **split 46:54, normal**; no furosemide needed.
- **Figure 7.2 — hydronephrosis without obstruction** (bilaterally repaired ureteroceles): less brisk right upstroke; delayed peak (11–14 min) and slow washout; **split 67:33 (abnormal)**. Obstructed systems usually keep rising through 30 minutes, so this pattern suggests a patulous system. **After 40 mg furosemide, $T_{1/2}$ 5 and 6.2 min → non-obstructed.**
- **Figure 7.3 — right obstruction:** better left perfusion (aortic curve unreliable because of scoliosis); small right kidney with a central photopenic dilated pelvis and **continuously rising curve**; **split 79:21**. After 40 mg furosemide, **no significant clearance from the obstructed right collecting system — a flat curve → obstruction confirmed.**

Additional Applications of ^{99m}Tc -MAG3 and ^{99m}Tc -DTPA Scintigraphy

Evaluation of Renal Vascular Hypertension

- **Mechanism:** autoregulation keeps GFR near-constant via **renin from juxtaglomerular cells**. Perceived low arteriolar flow → **renin: angiotensinogen → angiotensin I → angiotensin-converting enzyme (lung) → angiotensin II** → aldosterone release, **efferent arteriolar constriction** (maintaining filtration pressure) and peripheral vasoconstriction (hypertension).
- **Renal artery stenosis** keeps the renin-angiotensin-aldosterone system constantly active → hypertension and, in time, **glomerular sclerosis**.
- **Scintigraphy separates renovascular from essential hypertension. DTPA is preferred** because it is cleared purely by glomerular filtration.
- **Two-day protocol:** day 1, **oral captopril** (an angiotensin-converting enzyme inhibitor) → dynamic renogram; stenosis shows **slow uptake and low peak** → if abnormal, repeat 1–2 days later **off angiotensin-converting enzyme inhibitors and calcium channel blockers** → **improvement of 10% or more = high probability of renovascular hypertension** → likely benefit from angioplasty or surgery.
- **Why captopril unmasks stenosis:** the stenotic kidney maintains GFR through angiotensin II-mediated efferent constriction; blocking it drops filtration in that kidney.
- **Use has declined** with CT angiography, magnetic resonance angiography and Doppler ultrasonography.

Renal Transplant Evaluation

- MAG3 or DTPA helps separate **acute rejection** from **delayed graft function** (prolonged ischaemia at harvest), whose management differs. **Renal biopsy is the best differentiator**.

ACUTE REJECTION VERSUS DELAYED GRAFT FUNCTION ON DYNAMIC RENOGRAPHY

	Acute rejection	Delayed graft function
Perfusion	Decreased	Maintained
Serial studies	Worsens	Improves

- **Caveats:** overlapping causes of graft failure are common, and ischaemia can progress to chronic kidney disease, making serial imaging misleading — **clinical context is essential**.
- **Anastomotic (urine) leak:** tracer collecting outside the kidney on delayed images — valuable because transplant patients often **cannot receive iodinated contrast**.

Assessment of Vesicoureteral Reflux

- Vesicoureteral reflux (VUR) is retrograde flow from bladder to upper tracts, common in children, causing **ascending infection and eventual loss of renal function**.
- **Indirect method:** MAG3 or DTPA, then void once tracer is in the bladder. **Direct method (more common): ^{99m}Tc -sulfur colloid instilled into the bladder**.
- **Initial screening must be fluoroscopic voiding cystourethrography (VCUG)**, not scintigraphy: children with urinary tract infection also need assessment for **posterior urethral valves (males) and bladder diverticula**, which require anatomic detail.
- **Nuclear cystography is used for follow-up** — **more sensitive for subtle VUR, lower radiation dose** than contrast cystography.

FLUOROSCOPIC VERSUS NUCLEAR CYSTOGRAPHY FOR REFLUX

	Fluoroscopic VCUG	Nuclear cystography
Role	Initial screening	Follow-up

	Fluoroscopic VCUG	Nuclear cystography
Anatomy (valves, diverticula)	Yes	Little
Sensitivity for subtle VUR	Lower	Higher
Radiation	Higher	Lower

Renal Cortical Imaging With ^{99m}Tc -DMSA

- **Preparation as for dynamic imaging. Dose: 50 $\mu\text{Ci}/\text{kg}$ in children, 5 mCi in adults, intravenously → images at 2 hours with a pinhole collimator or SPECT. Uptake depends on filtration, so imaging may need delaying in renal failure.**
- **Normal:** homogeneous, smooth parenchyma. **Acute inflammation or infection:** photopenic defects. **Scarring:** also photopenic.
- **Serial imaging about 6 months apart separates the two:** pyelonephritic defects resolve; scars persist.

PYELONEPHRITIS VERSUS SCAR ON DMSA

	Acute pyelonephritis	Scar
Defect margins	Less defined	Sharper borders
Kidney	Normal size	Often small, atrophic; cortical thinning
Repeat at ~6 months	Resolves	Persists

KEY POINTS — FUNCTIONAL IMAGING OF THE KIDNEYS

- The commonest renal agents are ^{99m}Tc -DTPA, ^{99m}Tc -MAG3 and ^{99m}Tc -DMSA.
- **DTPA and MAG3** measure renal blood flow, differential function and obstruction.
- **DTPA is cleared by glomerular filtration; MAG3 by tubular secretion.**
- **MAG3 is preferred** at most centres: higher extraction efficiency, less affected by renal function.
- **DMSA is retained by proximal tubular cells** and is used for infection and scarring.
- DTPA and MAG3 also evaluate **renovascular hypertension, transplant function and VUR.**

Concept — Each renal tracer images the part of the nephron that handles it: filtration (DTPA) gives GFR, secretion (MAG3) gives the best dynamic curves whatever the GFR, and tubular retention (DMSA) gives a static map of the cortex.

INFECTION IMAGING

- Genitourinary infection is usually identified clinically and by culture; difficulty arises in **children, neurological disorders and immunocompromise.**
- **^{99m}Tc -DMSA is the most widely used agent for renal infection**, though its role has declined with CT. Active pyelonephritis → decreased uptake that resolves; **scar → persistent photopenia with cortical thinning.** Scars also occur in **VUR without documented pyelonephritis. Scarring predisposes to hypertension and chronic kidney disease.**
- **^{111}In -oxine leucocytes:** oxine is a lipid-soluble chelator that diffuses passively into leucocytes. **Sequence:** draw ~50 mL venous blood → remove erythrocytes → label leucocytes → reinject. **Normal uptake in liver, spleen and bone marrow;** uptake elsewhere means infection. Blood-pool clearance half-life ~7.5 h → **whole-body imaging at ~24 h.**

- **^{99m}Tc -hexamethylpropyleneamine oxime (HMPAO) leucocytes:** lipid-soluble neutral complex; **better image quality**, but **renal and hepatobiliary clearance obscures genitourinary infection**; image at **1–2 h** to limit bowel and urinary activity; **preferred in children** for dosimetry.
- **Transplant kidney: labelled leucocyte imaging cannot be used** — leucocytes accumulate in the graft, recognised as non-self.
- **Intra-abdominal infection** without localising symptoms or a convincing collection: **^{111}In -oxine leucocytes preferred**; intra-abdominal signal is **highly specific** for true infection.
- **^{67}Ga -citrate:** binds **transferrin** → carried to inflammation. Drawbacks: **non-specific uptake** in metabolically active sites, **delayed imaging to 48–72 h**, high cost.

INFECTION IMAGING AGENTS COMPARED

Agent	Mechanism	Imaging time	Strength	Limitation
^{99m}Tc -DMSA	Proximal tubular retention	2 h	Renal infection and scar	Displaced by CT
^{111}In -oxine leucocytes	Labelled autologous leucocytes	~24 h	Intra-abdominal; GU tract not obscured; highly specific	Not in transplant
^{99m}Tc -HMPAO leucocytes	Labelled autologous leucocytes	1–2 h	Image quality; paediatric dosimetry	Renal/hepatobiliary excretion obscures GU tract; not in transplant
^{67}Ga -citrate	Transferrin binding	48–72 h	Inflammation	Non-specific; slow; costly

Concept — *Leucocyte label choice is a trade between image quality and background:* ^{99m}Tc images better but is excreted through the very tract being examined, so ^{111}In is chosen for genitourinary and abdominal infection.

MOLECULAR IMAGING OF GENITOURINARY MALIGNANCIES

- **2-deoxy-2- ^{18}F fluoro-D-glucose (^{18}F -FDG)** is the commonest oncological PET tracer. **Sequence:** glucose analogue enters via **GLUT transporters** → **phosphorylated by hexokinase** → cannot diffuse out → lacking the 2-hydroxyl group it **cannot be metabolised further** → trapped and imaged.
- Physiological uptake in **brain and kidneys**. Malignant uptake reflects the **Warburg effect (aerobic glycolysis)** — a shift from oxidative phosphorylation to lactic acid fermentation.
- **FDG-imageable:** urothelial carcinoma, renal cell carcinoma (RCC), penile squamous cell carcinoma, testicular germ cell tumours. **Urinary excretion obscures the tract**, so FDG is reserved for **distant disease**.
- **Prostate cancer rarely uses aerobic glycolysis** → **poorly seen on FDG**.

Bladder Cancer

- **FDG is the most studied PET tracer** for bladder urothelial carcinoma (Fig. 7.4). For nodal staging a meta-analysis showed **modest sensitivity but high specificity**. PET **upstages** a significant minority thought to have organ-confined disease and changes management, including to palliation.
- **Guidelines do not support routine FDG-PET in clinically localised disease**, because outcome benefit is unproven. **American Urological Association (AUA):** PET only when conventional imaging is equivocal, or to confirm a finding that cannot be biopsied.

- **Upper tract urothelial carcinoma:** FDG-PET is most useful for distant disease; guidelines keep **conventional imaging for staging and follow-up**; PET selectively for patients at risk of metastatic recurrence who **cannot have contrast-enhanced imaging**.
- **Prognostication (emerging):** preoperative **maximum SUV** independently predicts recurrence after radical cystectomy with extended lymphadenectomy; FDG-PET may **predict histological response to induction or neoadjuvant chemotherapy**; **metabolic response** after systemic therapy then cystectomy is associated with lower progression and cancer-specific death in advanced disease.
- **^{11}C -choline and ^{11}C -acetate** (membrane building blocks, concentrating in rapidly dividing cells): potentially **more sensitive than FDG**, but **not approved**, and the **20.3-minute half-life** of ^{11}C has limited adoption.

Kidney Cancer

- **FDG-PET detects recurrent or metastatic RCC** with good sensitivity and specificity, but adds little beyond anatomic imaging. It is **most sensitive for type II papillary RCC**.
- **Guidelines recommend against routine FDG-PET for staging or follow-up of RCC**; moreover **clear cell RCC (ccRCC), the commonest subtype, is less FDG-avid** than other subtypes.
- **Carbonic anhydrase IX (CAIX):** a cell-surface protein **nearly universally expressed by ccRCC but not other histologies**. Leading agent: monoclonal antibody **girentuximab (G250)**.
- **^{124}I -girentuximab phase III trial:** for distinguishing localised ccRCC from other histologies, sensitivity **86.2%** and specificity **85.9%**, **outperforming contrast-enhanced CT** (sensitivity 75.5%, specificity 46.8%).
- **ZIRCON (confirmatory phase III)** uses ^{89}Zr -girentuximab (Fig. 7.5), because ^{89}Zr is "**charge-trapped**" in the tumour cytoplasm after antibody internalisation → **higher intratumoural signal, less blood-pool activity**. It is also being studied for nodal and distant ccRCC. At the time of writing, not yet approved.
- **$^{99\text{m}}\text{Tc}$ -sestamibi SPECT/CT (available off-label):** a **lipophilic cation binding cells rich in mitochondria**; approved for breast and myocardial imaging and widely used for **parathyroid adenoma**. It accumulates in **oncocyctic tumours** (high mitochondrial content, **low multidrug-resistance pump expression**) → **oncocytoma and hybrid oncocytic/chromophobe tumours are "hot"**, **ccRCC is "cold"** (Fig. 7.6). A 2024 meta-analysis showed high sensitivity and specificity, but the literature is small and adoption limited.

MOLECULAR IMAGING OF THE RENAL MASS

Agent	Target	Hot lesion	Status
^{18}F -FDG PET	Glucose metabolism	Type II papillary RCC most; ccRCC less avid	Not routine (guidelines against)
$^{89}\text{Zr}/^{124}\text{I}$ -girentuximab PET	CAIX	Clear cell RCC	Investigational (ZIRCON)
$^{99\text{m}}\text{Tc}$ -sestamibi SPECT/CT	Mitochondria	Oncocytoma, hybrid oncocytic/chromophobe	Off-label; limited adoption

- **Sestamibi and girentuximab are mirror images:** sestamibi lights up the benign oncocytic tumour; girentuximab lights up clear cell carcinoma.

Prostate Cancer

- Historically: **contrast CT, MRI and bone scintigraphy with $^{99\text{m}}\text{Tc}$ -methylene diphosphonate (MDP)**. Now increasingly **PET targeting prostate-specific membrane antigen (PSMA)**.

- **National Comprehensive Cancer Network (NCCN, version 4.2024): PSMA-PET is the preferred imaging method;** conventional imaging is **not a prerequisite**, and PSMA-PET/CT or PET/MRI is an equally or more effective **front-line tool at initial staging and biochemical recurrence**.
- **¹¹C-choline:** added value for **occult pelvic nodes** before radical prostatectomy; in biochemical recurrence, **detection rises with prostate-specific antigen (PSA)**. Limited by the **on-site cyclotron and 20.3-minute half-life**. Fluorinated derivatives (¹⁸F-fluorocholine, ¹⁸F-fluoromethylcholine) are **not approved in most countries**, including the USA.
- **¹⁸F-fluciclovine (¹⁸F-FACBC):** synthetic amino acid; substrate for **LAT1 and ASCT2** transporters; **¹⁸F half-life 109.7 min** allowed wider use. Performance parallels choline. **Both metabolic tracers take up non-specifically in inflammation, infection and benign prostatic hyperplasia (BPH)** → poor for local prostate imaging.
- **PSMA: type II transmembrane glycoprotein** highly expressed by prostate cancer epithelium; **expression rises with grade, stage and recurrence risk**. Limited normal expression — **salivary and lacrimal glands, jejunum, neuronal ganglia**.
- **Agents:** early work with **⁶⁸Ga-PSMA-11 and ⁶⁸Ga-PSMA-I&T**; trend to **¹⁸F agents** (DCFPyL, rhPSMA-7.3, PSMA-1007) for longer half-life and better images.

US FOOD AND DRUG ADMINISTRATION (FDA)-APPROVED PSMA PET AGENTS

Agent	Other names
⁶⁸ Ga-PSMA-11	Kits: LOCAMETZ, ILLUCCIX
¹⁸ F-DCFPyL	Piflufolastat F 18, PYLARIFY
¹⁸ F-rhPSMA-7.3	Flotufolastat F 18, POSLUMA

- **PSMA beats metabolic tracers in sensitivity and specificity** — shown in the **landmark Calais trial (2019)**: in biochemical recurrence, **patient-level detection 26% with ¹⁸F-FACBC vs 56% with ⁶⁸Ga-PSMA-11**. A retrospective comparison with ¹¹C-choline agreed; specificity **exceeds 90%** across studies.
- **Primary staging** of men at risk of nodal or metastatic disease (Fig. 7.7): prospective trials of PSMA-PET before prostatectomy with pelvic lymphadenectomy showed **nodal sensitivity 30–40%, specificity above 90%** — **a positive node is reliable; a negative scan does not exclude micrometastases**.
- **Biochemical recurrence:** prospective trials show excellent detection **even at very low PSA**. **CONDOR (¹⁸F-DCFPyL): detection 61.5% at median PSA 0.8 ng/mL; management changed in 63.9%** — about four-fifths of changes followed positive scans and one-fifth negative scans.
- **Outcome evidence: no data yet show that PSMA-driven management changes improve outcomes**. **EMPIRE-1 (¹⁸F-FACBC):** after prostatectomy, salvage radiotherapy directed by FACBC-PET vs conventional imaging → plan changed in ~35% (e.g. adding pelvic nodes) and **12.5% absolute improvement** in 3-year freedom from the composite of PSA failure, imaging progression and systemic therapy. A PSMA equivalent is under way.
- **Oligometastatic disease (not yet guideline-supported): ORIOLE (phase II)** — metastasis-directed radiotherapy chosen on conventional imaging, with blinded ¹⁸F-DCFPyL; **untreated PSMA-avid disease → progression 63% vs 16%** at 6 months when all avid sites were treated.
- **Radioligand therapy: VISION (2021)** → **¹⁷⁷Lu-PSMA-617** (beta emitter) approved for **metastatic castration-resistant prostate cancer after androgen receptor pathway inhibitor and taxane**. **PSMA-PET is essential to confirm target expression**, which can be lost with **neuroendocrine transdifferentiation**.
- **Figure 7.7:** a left iliac lesion near the sacroiliac joint seen on MDP bone scan, sclerotic on CT and **PSMA-avid on ¹⁸F-DCFPyL** — concordant metastasis.

MOLECULAR IMAGING TRIALS IN PROSTATE CANCER

Trial	Tracer	Setting	Key result
Calais (2019)	⁶⁸ Ga-PSMA-11 vs ¹⁸ F-FACBC	Biochemical recurrence	Detection 56% vs 26%
CONDOR	¹⁸ F-DCFPyL	Biochemical recurrence	61.5% detection at PSA 0.8; 63.9% management change
EMPIRE-1	¹⁸ F-FACBC	Salvage radiotherapy after prostatectomy	12.5% absolute gain in 3-year event-free survival
ORIOLE	¹⁸ F-DCFPyL (blinded)	Oligometastatic, metastasis-directed radiotherapy	Progression 63% vs 16% (untreated vs all avid sites treated)
VISION (2021)	¹⁷⁷ Lu-PSMA-617 (therapy)	mCRPC after ARPI and taxane	Approval of radioligand therapy

PROSTATE PET TRACERS COMPARED

	¹¹ C-choline	¹⁸ F-FACBC (fluciclovine)	PSMA agents
Mechanism	Membrane synthesis	Amino acid transport (LAT1, ASCT2)	Cell-surface PSMA
Half-life	20.3 min (on-site cyclotron)	109.7 min	68 min (⁶⁸ Ga); 110 min (¹⁸ F)
Specificity	Limited (inflammation, BPH)	Limited (inflammation, BPH)	>90%
Status	Limited adoption	Approved; EMPIRE-1 outcome data	Preferred (NCCN)

Penile Cancer

- **Penile squamous cell carcinoma is highly FDG-avid.**
- **Palpable inguinal nodes:** FDG-PET reliably **confirms metastasis and its extent** (very high sensitivity on meta-analysis).
- **PET-positive nodal disease carries worse overall survival**, and **only PET-positive patients benefited from neoadjuvant chemotherapy** → FDG-PET helps **steer PET-positive patients to neoadjuvant chemotherapy**.
- **Limited role: non-palpable nodes** (poor sensitivity for occult metastases) and **primary tumour staging**; data for distant disease are scarce.

Testis Cancer

- **Seminoma:** European Association of Urology (EAU) endorses FDG-PET for a residual post-chemotherapy mass >3 cm, to separate **fibrosis from viable tumour**; meta-analysis showed moderate-to-good sensitivity and specificity.
- **Non-seminomatous germ cell tumour (NSGCT):** FDG-PET **not indicated** — no benefit over CT plus serum markers. Standard of care is **retroperitoneal lymph node dissection for residual masses >1 cm**; a PET tracer for viable NSGCT is a critical unmet need.

POST-CHEMOTHERAPY RESIDUAL MASS: SEMINOMA VERSUS NON-SEMINOMA

	Seminoma	Non-seminomatous germ cell tumour
FDG-PET	Indicated if residual mass >3 cm	Not indicated
Question answered	Fibrosis vs viable tumour	— (CT + markers)
Surgical threshold	—	RPLND if >1 cm

- **Why:** teratoma, common in NSGCT residual masses, is not FDG-avid, so PET cannot exclude it.

KEY POINTS — MOLECULAR IMAGING

- Molecular cancer imaging most commonly uses **^{18}F -FDG PET**.
- Several genitourinary malignancies are FDG-imageable, with **varying utility beyond anatomic imaging**.
- **FDG is excreted in urine**, so it is used mainly for **distant disease**.
- **FDG has little role in prostate cancer**; other tracers were developed for it.
- **PSMA-targeted PET has become the standard of care** for prostate cancer imaging in many parts of the world.
- A well-established FDG-PET indication is **residual seminoma after chemotherapy**.

Concept — A tracer is only as useful as the biology it exploits: FDG follows glycolysis, so it fails in indolent prostate cancer and clear cell RCC and is swamped by urinary excretion, whereas target-specific agents (PSMA, CAIX, mitochondrial content) succeed because they image a molecule the tumour actually carries.